r/NoStupidQuestions

How much energy does asking ChapGPT a question consume versus using Google and their AI providing an answer?

Posted by u/growtreesbreathelife • • 0 points • 2 comments

I recently came across a post in a baking subreddit in which someone was asking to scale a recipe, I suggested as a tool to use ChatGPT to quickly get the answer as it's usage had aided me when I was in a hurried pinch. I got flamed hard, people telling me that I shouldn't use it to ask it such simple questions because it uses a lot of energy and consumes a lot of water so I've kind of been avoiding it but I just made a search on Google and got a google AI overview and wondered if Google calculating an AI response would consume around the same energy as a ChatGPT inquiry

Comments

u/archpawn • 1 points • 2025-06-03 19:02 UTC

Google is in the business of making money. The AI doesn't just have to use a small enough amount of electricity for the ad revenue to pay for it. It has to use a small enough amount that the ad profit from the extra people using the service because of the AI overview to pay for it. Also, the AI overview doesn't have to remember anything like ChatGPT does. It just needs to summarize the results its given, and can use a much smaller model. It also tends to give much shorter responses than ChatGPT.

I've heard ChatGPT uses about ten times as much electricity as a a google search. Given everything, I'd guess the AI overview uses less electricity than the rest of the search.

u/Huck_N_Fell • 1 points • 2025-06-03 18:58 UTC

Here is the response from ChatGPT…

The energy usage for answering a single question on ChatGPT versus doing a Google search can vary depending on many factors, but here’s a general comparison based on recent estimates:

🔷 ChatGPT (Large Language Model Inference)
• Estimated energy per query: ~2.9 to 4.0 watt-hours (Wh)
• This is based on data from a 2023 study by researchers at Hugging Face and other sources, depending on model size and hardware (e.g., GPUs like A100s).
• The energy used can be higher for longer or more complex prompts.

🔷 Google Search
• Estimated energy per query: ~0.3 watt-hours (Wh)

• Google has made its data centers extremely efficient. Public data from Google suggests their search infrastructure is highly optimized compared to LLM inference.

🔍 Why the Difference?
• LLMs like ChatGPT use huge transformer models with billions of parameters, which require more GPU compute per query.
• Search engines use indexed data and lightweight ranking algorithms, which are computationally cheaper.

⚙️ Efficiency Notes
• OpenAI and other companies are working on making models more energy-efficient through better hardware, model distillation, and infrastructure improvements.
• The energy per query can vary based on the deployment (e.g., edge device, cloud data center, size of the model — GPT-4 vs GPT-3.5, etc.).

u/Delehal • 1 points • 2025-06-03 18:51 UTC

Google Search, minus the AI overview, is quite efficient. Once you add in the AI overview, though, that's using a large language model (LLM) just like ChatGPT does. I haven't seen any articles that directly compare those two AI services, but it seems reasonable to assume that they would be ballpark similar ish.